

What Is a Polar Vortex Weather System?

Weather experts say extreme cold in many parts of the United States this winter has been driven by a polar vortex. But what is this weather system and how does it work?

The National Oceanic and Atmospheric Administration (NOAA) describes a polar vortex as strong westerly winds that form each winter in the atmosphere above the North Pole.

These cold winds move about 16 to 48 kilometers above the surface in the Arctic area in the part of the atmosphere called the stratosphere.

At times, the polar vortex can weaken, causing masses of cold air to break off and move south. A polar vortex can also push warmer air in northerly directions.

The Associated Press (AP) reports that several vortex events have affected parts of the U.S. this winter. Experts say weather forces in the Arctic have combined to push colder air to the U.S. and Europe.

Temperature readings that take into account the effects of wind, called windchill readings, fell to minus 51 degrees Celsius in parts of the U.S. Midwest. Meteorologists say windchill readings are colder than air temperature readings.

The U.S. National Weather Service (NWS) warned people in affected areas that such low temperatures can cause frostbite and hypothermia if skin is not protected from the cold. Weather officials also warned that long-term cold temperatures can cause water pipes in or around homes to freeze and burst.

The NWS noted that almost every U.S. state – except for Hawaii, California and Florida – is expected to report low windchill temperatures. The coldest temperatures were expected in the midwestern states of Kansas, Nebraska and Missouri.

In an explanation on its website, NOAA states that the polar vortex always surrounds both of Earth's poles. The systems strengthen in the winter and weaken in the summer.

Every year or so, the polar vortex becomes weaker. This can cause the system to be “pushed off the pole or split into two.” This can lead to cold air at the surface being pushed southward and warm air being pushed to the north.

NOAA says its weather models are predicting low temperatures for states including Oklahoma, Colorado, Nebraska, Missouri, Illinois and Iowa. Other states likely to have very low temperatures are Wyoming, Montana, North Dakota, South Dakota, Minnesota, Wisconsin and Michigan.

The agency said some storms could bring heavy snow or flooding during that time.

I'm Bryan Lynn.

Bryan Lynn wrote this story for VOA Learning English, based on reports from The Associated Press, Reuters, Agence France-Presse, NOAA and NASA.

Vocabulary

northerly - adj. ทางทิศเหนือ, ไปทางทิศเหนือ, มาจากทางทิศเหนือ

frostbite - n. อาการเนื้อเยื่อถูกทำลายจากความเย็นจัด (เช่น นิ้วมือ นิ้วเท้า)

hypothermia - n. ภาวะอุณหภูมิร่างกายต่ำกว่าปกติ, ภาวะตัวเย็นเกิน (เป็นภาวะร้ายแรงที่ร่างกายสูญเสียความร้อนเร็วกว่าที่สร้างได้)

windchill - n. อุณหภูมิที่รู้สึกหนาวเย็น (เมื่อลมพัด), ดัชนีความหนาวเย็นจากลม (เป็นค่าที่คำนวณจากอุณหภูมิอากาศจริงและความเร็วลม เพื่อแสดงให้เห็นว่าลมทำให้ร่างกายรู้สึกหนาวเย็นกว่าอุณหภูมิจริงเท่าใด)

Reading Comprehension Quiz

1. What is identified as the primary cause of extreme cold in many parts of the United States this winter?

- Global warming
- A polar vortex
- Unusual ocean currents
- High-pressure systems

2. According to the National Oceanic and Atmospheric Administration (NOAA), what is a polar vortex characterized as?

- A tropical storm system
- Strong westerly winds
- A high-pressure zone
- A layer of warm air

3. Where does a polar vortex typically form each winter?

- Above the equator
- In the stratosphere above the North Pole
- At sea level in the Arctic
- In the Southern Hemisphere

4. At what approximate altitude do the cold winds of the polar vortex move above the surface?

- 1 to 5 kilometers
- 16 to 48 kilometers
- 50 to 100 kilometers
- Below 1 kilometer

5. In which part of the atmosphere are the polar vortex winds specifically located?

- The troposphere
- The mesosphere
- The stratosphere
- The exosphere

6. What is one consequence when the polar vortex weakens?

- It causes tropical storms to form.
- It pushes warmer air towards the equator.
- It leads to cold air breaking off and moving south.
- It strengthens the jet stream.

7. Besides pushing cold air south, what other effect can a polar vortex have on air movement?

- It can pull warm air downwards.
- It can push warmer air in northerly directions.
- It can create a vacuum effect.
- It can stabilize atmospheric pressure.

8. Which news agency reported that several vortex events have affected parts of the U.S. this winter?

- Reuters
- Agence France-Presse
- The Associated Press (AP)
- VOA Learning English

9. Experts attribute the push of colder air to the U.S. and Europe to what phenomenon?

- Increased solar activity
- Deforestation in the Arctic
- Weather forces in the Arctic combined

Volcanic eruptions

10. How do meteorologists describe windchill readings compared to air temperature readings?

They are usually warmer.

They are identical.

They are colder than air temperature readings.

They are only relevant in tropical climates.

11. What was the lowest windchill reading reported in parts of the U.S. Midwest?

Minus 15 degrees Celsius

Minus 51 degrees Celsius

Plus 20 degrees Celsius

Zero degrees Celsius

12. According to the U.S. National Weather Service (NWS), what are two potential health risks if skin is not protected in very low temperatures?

Heatstroke and sunburn

Dehydration and heat exhaustion

Frostbite and hypothermia

Allergies and rashes

13. What type of property damage can long-term cold temperatures lead to, according to weather officials?

Roof collapse

Electrical outages

Water pipes freezing and bursting

Foundation cracks

14. Which three U.S. states were specifically mentioned as *not* expected to report low windchill temperatures?

New York, Texas, and Washington

Hawaii, California, and Florida

Alaska, Maine, and Vermont

Colorado, Illinois, and Iowa

15. Which three midwestern states were predicted to experience the coldest temperatures?

Ohio, Indiana, and Michigan

North Dakota, South Dakota, and Minnesota

Kansas, Nebraska, and Missouri

Wisconsin, Illinois, and Iowa